

0

DATA BASE SYSTEMS ARCHITECTURE

Databases III

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UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

- 1 Database System Administration
- 2 Record Storage
- 3 DBMS Architecture
- 4 Transactional System
- 5 Query Execution
- 6 Concurrency Control
- 7 Failure Recovery



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Database System Administration

- **Database system administration** is the discipline of **managing**, **configuring**, and **maintaining database systems** to ensure their **reliability**, **performance**, **security**, and **availability**.

- Key responsibilities include:

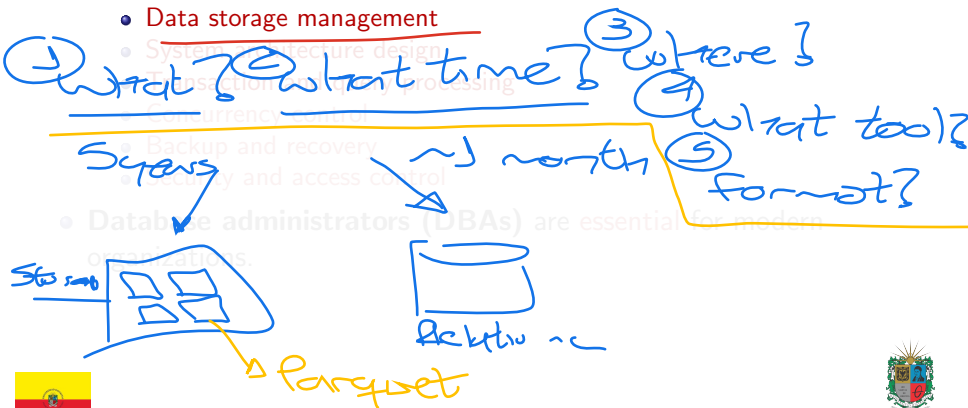


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 - Data storage management
 - System architecture design

physical
infrastructure

on-premise? cloud? hybrid?

logical architecture → SQL? NoSQL?
 minors? sync? scaling?
 etl-elt? data lake?
 data warehouse? data mart?



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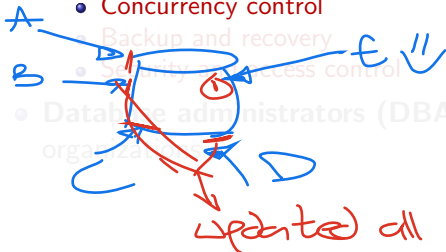
tool → failure
!!
☹

response time? concurrency?
order in query? optimization?



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- Security and access control

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fault tolerant?

copy-availability

self-recovery?



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Database administrators (DBAs) are essential for modern applications.

SSH → Grants? → 2FA → auth 2 authenticator

↓ protocols



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Record Storage Concepts

- A **record** (or row/tuple) is the basic unit of data storage in a database table.

- Efficient record storage is critical for fast data retrieval and update.
- Storage techniques:
 - Heap storage: fast insertion but slow retrieval.
 - B-tree storage: fast retrieval but slow insertion.



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- **Efficient record storage** is crucial for **fast data retrieval** and **update**.

• Storage techniques:

SELECT
self)

1. Operation with
most executions

2. Cost $\rightarrow$ money
 $\rightarrow$ space

Create
Update
Delete
save (row)



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- Indexed storage: Uses indexes for fast lookups.

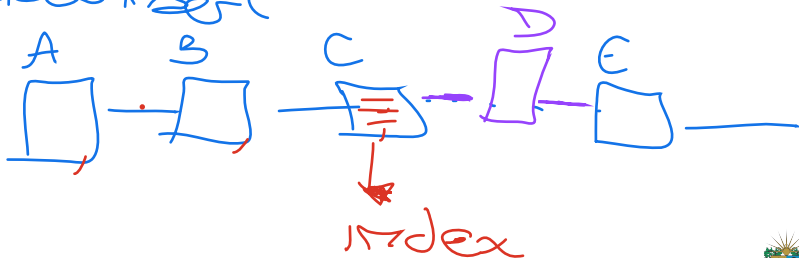


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slow insert



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Altra skua updates

A {1-~~25~~2} → 12

B {~~26-23~~27-35}

C {~~3-3-4~~34-46}



Block and Page Organization

- Data is stored in **blocks** (pages) on disk.

- Block size and layout affect I/O performance.

- Records may be packed, split or span multiple blocks.

- Free space management is important for updates and inserts.

- Fragmentation management ensures efficient use of space.

- Data compression can also improve storage efficiency.



video



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+ performance

+ storage

+ hard to migrate



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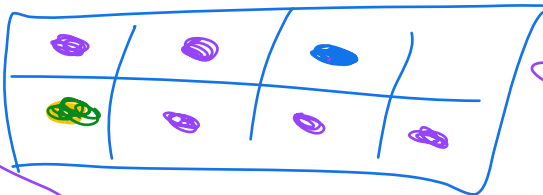
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2GB → J.S.E.
Free



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defragmented



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|
- redundant
+ Fast load

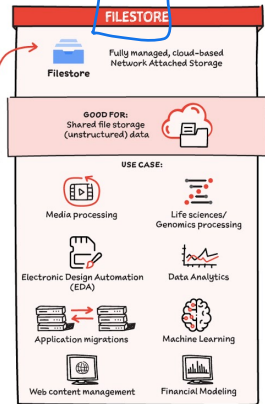
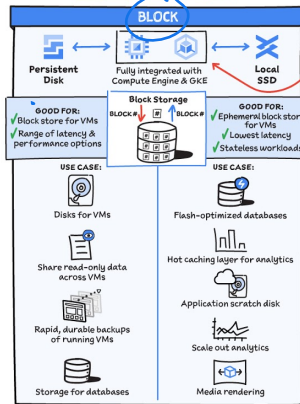
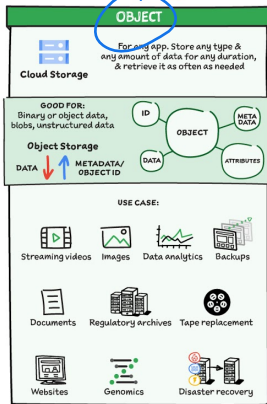


Record Storage: Image

#GCPsketchnote
 @PVERGADIA
 THECLOUDGIRL.DEV
 04.23.2021



Which Storage Should I Use?



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DBMS Architecture Overview

- Engines*
- A **Database Management System (DBMS)** is organized in **layers**:
 - Storage Manager**: Handles data storage, file organization, and access methods.



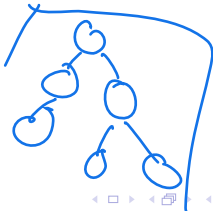
- Each component is responsible for a **specific aspect** of data management.



- Declarative $\rightarrow$ what

→ **SELECT** → 1 row
for row in table A:
data.append(row)

Every



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- A **Database Management System (DBMS)** is organized in **layers**:

- **Storage Manager**: Handles *data storage*, *file organization*, and *access methods*.
- **Query Processor**: Parses, optimizes, and executes *SQL queries*.
- **Transaction Manager**: Ensures *ACID properties* for transactions.

- **Concurrency Control Manager**: Manages *simultaneous operations* and prevents conflicts.

- **Recovery Manager**: Handles *failures* and *restores data consistency*.

- **Query Optimizer**: Selects the most efficient strategy.

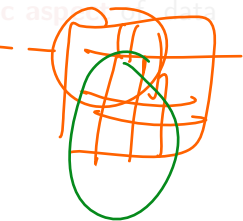
Atomicity → Debug, understanding
Consistency.

Isolated

Durability

Queue request

execute



DBMS Architecture Overview

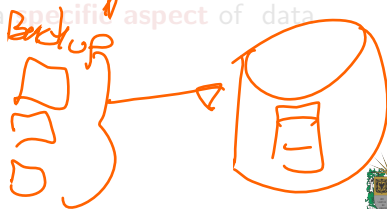
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if I ~ bbbkey &
 then restart
 com A → B
 or crypt



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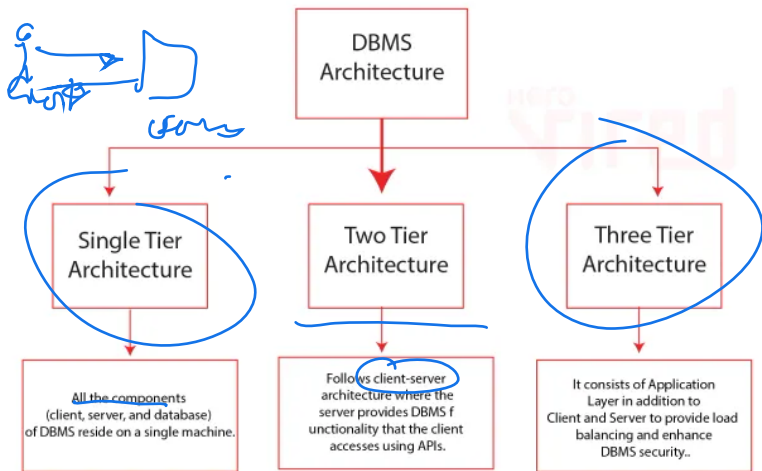


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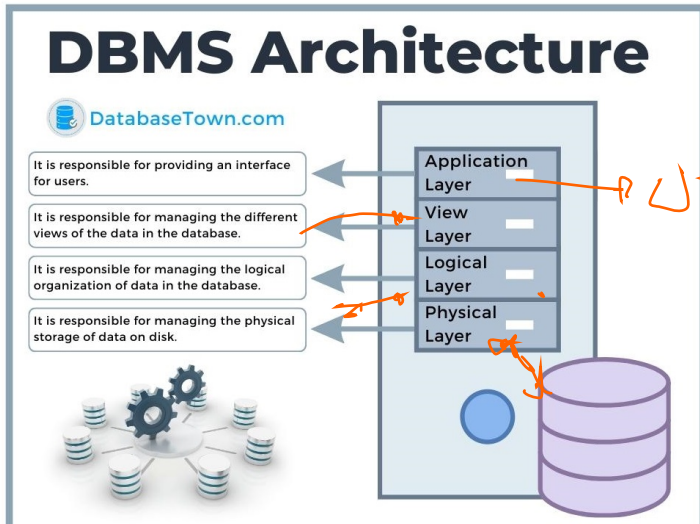
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DBMS Architecture Tiers



DBMS Architecture N-Tier



Types of DBMS Architecture

There are several types of DBMS architectures:

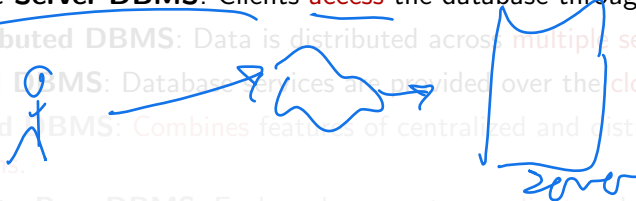
- Centralized DBMS: All components are on a **single server**.
- Client-Server DBMS: Clients **access** the database through a **server**.
- Distributed DBMS: Data is distributed across **multiple servers**.
- Cloud DBMS: Database services are provided over the **cloud**.
- Hybrid DBMS: **Combines** features of centralized and distributed systems.
- Peer-to-Peer DBMS: Each node can act as a **client** and **server**.
- In memory DBMS: Data is stored in **RAM** for faster access.



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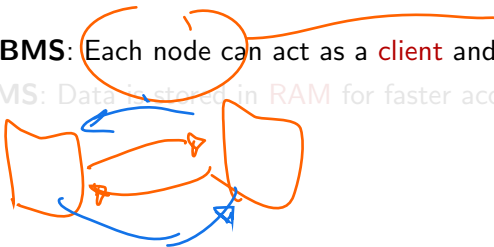
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Cadé
↳ Redis



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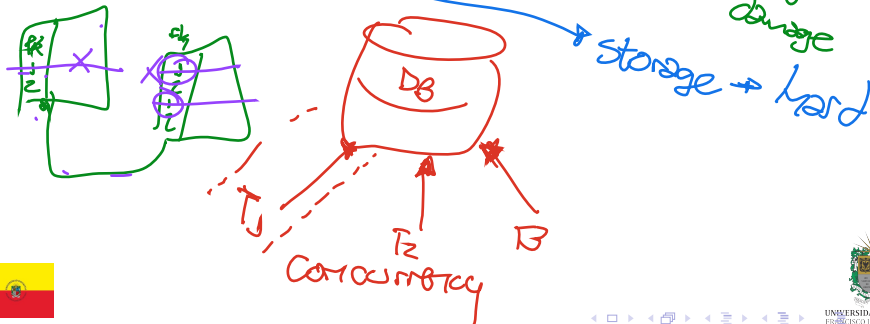


Transactional System Concepts

- A **transaction** is a sequence of operations performed as a **single** logical unit of work. *consulta → query*

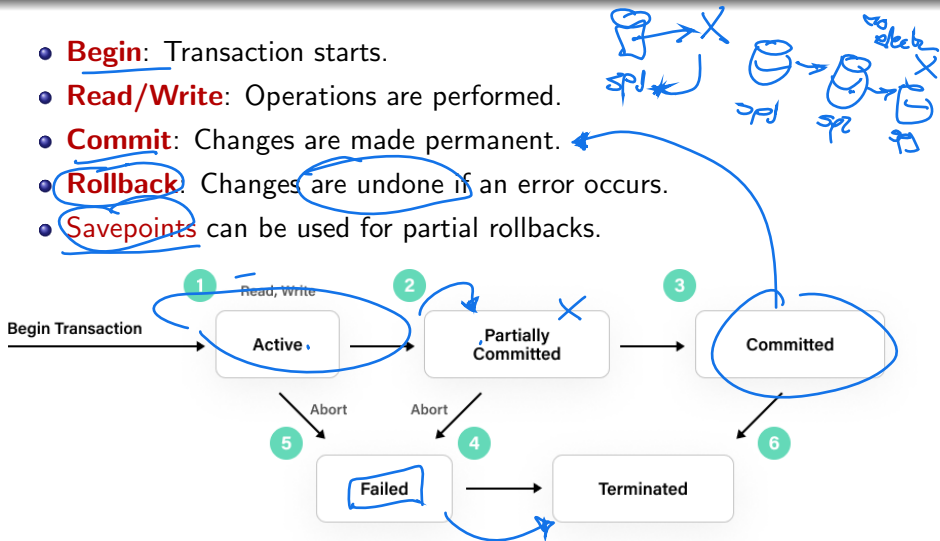
- Transactions must satisfy the **ACID** properties:

- **Atomicity**: All or nothing. *→ execution*
- **Consistency**: Preserves database integrity.
- **Isolation**: Transactions do not interfere.
- **Durability**: Results persist after completion. *→ storage → hard*



Transaction Lifecycle

- **Begin:** Transaction starts.
- **Read/Write:** Operations are performed.
- **Commit:** Changes are made permanent.
- **Rollback:** Changes are undone if an error occurs.
- **Savepoints** can be used for partial rollbacks.



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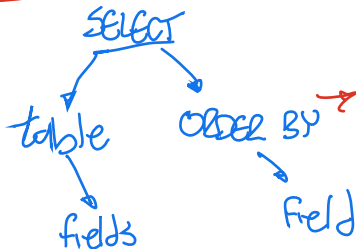
Query Execution Process

- **Query execution** is the process of interpreting and running database queries.

Steps:

- Parsing: Analyzing query syntax.
- Optimization: Choosing the best execution plan.
- Execution: Retrieving and processing data.

- Efficient execution is critical for performance.



~~INTO INSERT~~
INSERT INTO

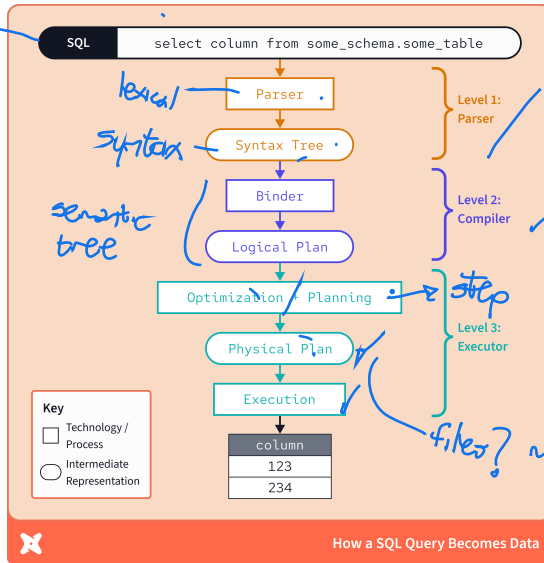
how? → engine

for (_)
if (_)

words
lexical
syntax → order



Query Execution Flow: Full Transaction



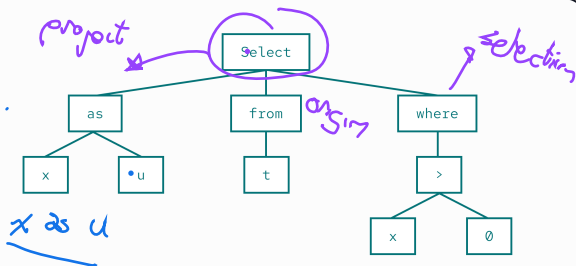
Query Execution Flow: Syntax Tree

SQL

select x as u from t where x > 0

Parser

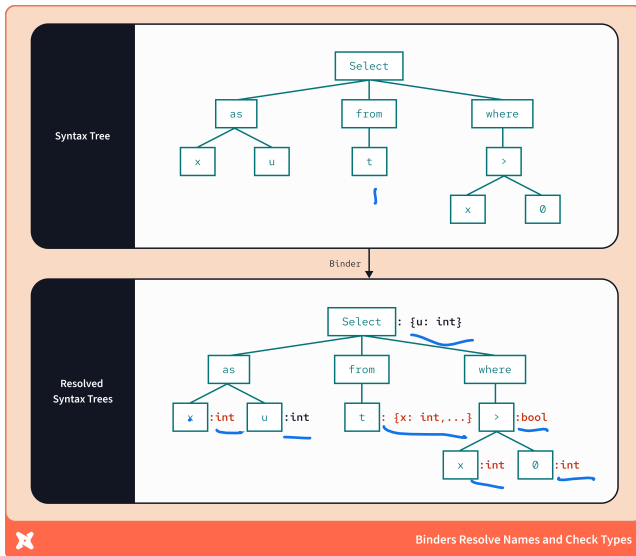
Syntax Tree



Parsers Recognize the Structure of the Query

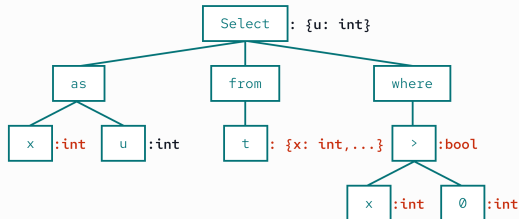


Query Execution Flow: Compilation



Query Execution Flow: Logical Plan

Resolved
Syntax Trees



Generating Logical Plan

Logical Plan

1. TableScan: t *→ for*
2. Filter: t.x > 0 *→ condn } filter*
3. Projection: t.x as u *→ for x cond*



Compilation Produces an Executable Plan from a Resolved Syntax Tree



Query Optimization

- The query optimizer selects the most efficient strategy for executing a query.
 - Considers indexes, join methods, and data distribution.
 - May *re-write* queries for better performance.
 - Cost-based and rule-based optimization approaches.
- Handwritten notes:*
- memory
 - processor
 - amount & order
- Handwritten diagram:* A blue line under "strategy" with an arrow pointing down to the word "Plan" written in blue.



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column
 A J=10
 B X 11-15
 C 16-20
 ...

data



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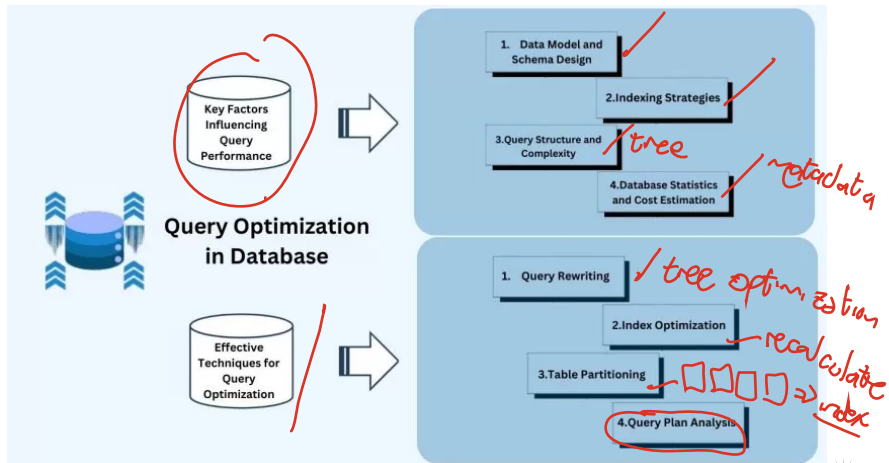


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Query Optimization Factors



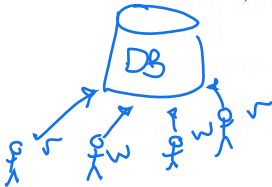
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- Prevents problems like lost updates, dirty reads, and deadlocks.
- Techniques:



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- Techniques:
 - **Locking protocols**: Use locks to control access (e.g., two-phase locking).

- **Timestamp ordering**: Assigns timestamps to transactions.
- **Optimistic concurrency control**: Assumes no conflicts.
- **Pessimistic concurrency control**: Assumes conflicts may occur.

Consistency + Integrity

Write → lock read
read → lock write

- waiting time
increase
- memory due
queues



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↳ T₁: J:J.00J
 T₂: J:J.00J
 Dirty Reads
 Fast



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→ Parallel without locks
 ↳ reduce queues
 ↳ memory

high response time



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- locking + timestamp



Locking and Deadlocks

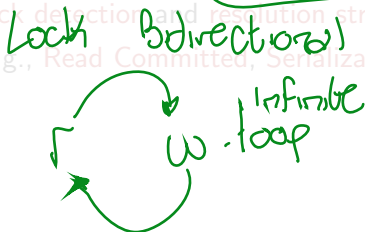
W. → 85
r. → 90

- **Locks** prevent conflicting operations on the same data.
- Deadlocks occur when transactions wait indefinitely for each other.
- DBMS uses deadlock detection and resolution strategies.
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↓
loops?



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ACID



Outline

- 1 Database System Administration
- 2 Record Storage
- 3 DBMS Architecture
- 4 Transactional System
- 5 Query Execution
- 6 Concurrency Control
- 7 Failure Recovery**



Failure Recovery Concepts

- **Failure recovery** restores the database to a **consistent state** after a failure.

- Types of failures:

- Transaction failures: one transaction fails.
- System failures: all transactions are lost.
- Media failures: for storage device fails.



- Recovery techniques:

- Write-ahead logging (WAL)
- Last stable version
- Metadata

last → good health



Failure Recovery Concepts

- **Failure recovery** restores the database to a **consistent state** after a failure.

- **Types of failures:**

- **Transaction failure:** Only one transaction fails.
- **System crash:** All active transactions are lost.
- **Media failure:** Disk or storage device fails.

- **Recovery techniques:**

- Write-ahead logging (WAL):

- Checkpointing:

domino effect

→ process close
↓
ACID



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- Recovery techniques:
 - **Write-ahead logging (WAL)**:
 - Logs changes before applying them.
 - Ensures durability and atomicity.



Plain-text

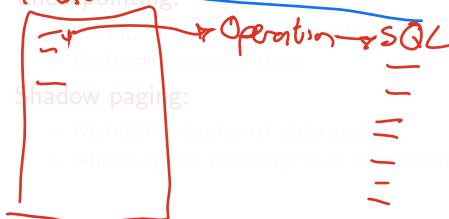
Shadow paging:

Reduces recovery time.

Shadow paging:

Maintains copies of data pages.

Allows quick recovery to a consistent state.



{ how/when user action
 CREATE
 UPDATE
 DELETE



Failure Recovery Concepts

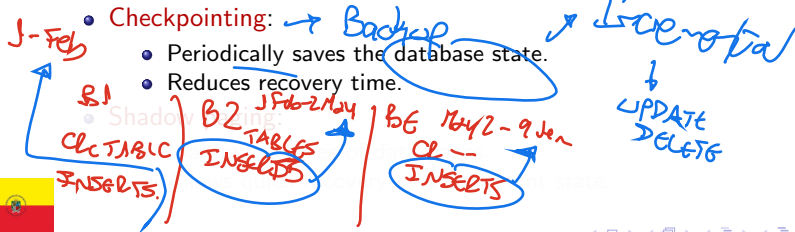
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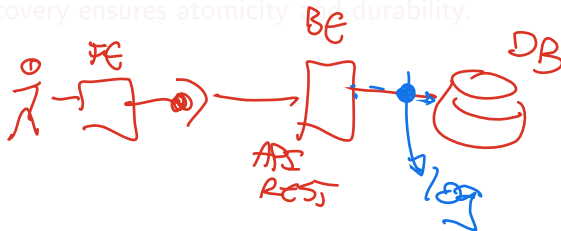
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Write-Ahead Logging and Checkpoints

- **Write-ahead logging (WAL):** All changes are **logged before** being applied to the database.
- **Checkpoint** *- storage*: Periodically save the database state to speed up recovery. *+ format*
- **Shadow paging** *ACID*: Maintains copies of data pages. *→ Open in 2ry tech*
- **Recovery** ensures atomicity and durability.



Write-Ahead Logging and Checkpoints

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- **Shadow paging:** Maintains **copies of data pages**.
- **Recovery** ensures atomicity and durability.

Automatic

1 month
1 day - 1 hour
10 minutes



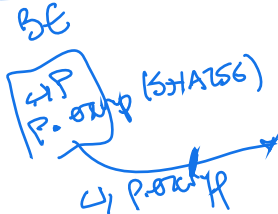
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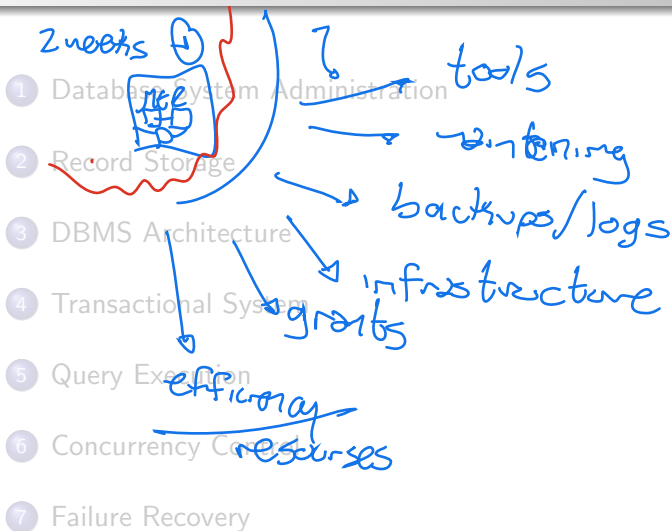


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Outline



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/databases-ii>

